

AUGUST/RESIT EXAMINATIONS 2021/2022

MODULE: CA267 - Software Testing

PROGRAMME(S):
CASE BSc in Computer Applications (Sft.Eng.)

YEAR OF STUDY: 2

EXAMINER(S): Dr. Silvana MacMahon (Internal) (Ext:5620)

TIME ALLOWED: 2 Hours

INSTRUCTIONS: Answer all questions.

PLEASE DO NOT TURN OVER THIS PAGE UNTIL YOU ARE INSTRUCTED TO DO SO.
The use of programmable or text storing calculators is expressly forbidden.

There are no additional requirements for this paper.

QUESTION 1**[TOTAL MARKS: 25]****Q 1(a)****[5 Marks]**

Provide a description of functional testing and non-functional testing, identifying an example of each type of testing.

Q 1(b)**[10 Marks]**

Describe five principles of software testing, explaining why each principle is of relevance to software testing.

Q 1(c)**[10 Marks]**

Explain the difference between System Testing and Acceptance Testing Levels. Use examples to identify issues that might arise in acceptance testing but which were not evident in system testing.

[End of Question 1]**QUESTION 2****[TOTAL MARKS: 25]****Q 2(a)****[10 Marks]**

Provide an example of a basic software requirement (other than those provided in the course materials), and outline an approach to testing the quality of that requirement. Explain the challenges associated with requirement definition and why requirements need to be tested.

Q 2(b)**[7 Marks]**

Distinguish between scripted testing and low formality testing techniques, using examples to illustrate your answer. Explain what factors you may need to consider if using exploratory testing on a project.

Q 2(c)**[8 Marks]**

What is the difference between debugging and testing? In your opinion, which is more important. Use examples to illustrate your answer.

[End of Question 2]

QUESTION 3**[TOTAL MARKS: 25]****Q 3(a)****[5 Marks]**

What are software testing exit criteria, and in your opinion, are they necessary? Using specific examples. identify circumstances where exit criteria might be relaxed.

Q 3(b)**[7 Marks]**

Identify and briefly explain five distinct ways in which poor configuration management might affect software testing. In your opinion which one these should be addressed first and why?

Q 3(c)**[6 Marks]**

Explain, using examples to illustrate, what is meant by "Test Driven Development or TDD". Would you advocate the use of TDD? Explain your rationale clearly.

Q 3(d)**[7 Marks]**

What are stubs and drivers, why are they produced in some testing environments, and, in your opinion, is the cost associated with their creation and maintenance warranted?

[End of Question 3]**QUESTION 4****[TOTAL MARKS: 25]****Q 4(a)****[7 Marks]**

You are the driver in a Pair Programming team and the navigator on your team is not engaged with the process. Explain using examples the impact that this will have on the potential benefits of using Pair Programming. In your opinion, what would be the best ways to address this issue and why?

Q 4(b)**[8 Marks]**

Explain the use of control flow graphs. Describe clearly using examples to illustrate your point how these graphs can be used for testing.

Q 4(c)

[10 Marks]

"Exploratory Testing has a bad reputation in some circles".

Explain using examples why you think that this may be the case. In your opinion, is this reputation justified and explain if you would advocate this approach or not. (State any assumptions that you have made)

[End of Question 4]

[END OF EXAM]